What`s new in version 4.27.21.005

1. Parameter P(33) was added – total pressure at first stage of LPT discharge

2. Parameters T(33) and G(33) were renamed to temperature and gas flow at first stage of LPT discharge.

3. Error of file saving with control laws was fixed.

4. Error of data importing from Excel was fixed.

What`s new in version 4.27.21.002

1. New version has an editor to input the adjustments to the performances of LPT ($\Delta G = f(Re)$, $\Delta\eta = f(Re)$). To call the editor User must choose file in menu ***Add files Reinolds - The performance that considers Reynolds number effect on LPT performances*** (fig. 1)

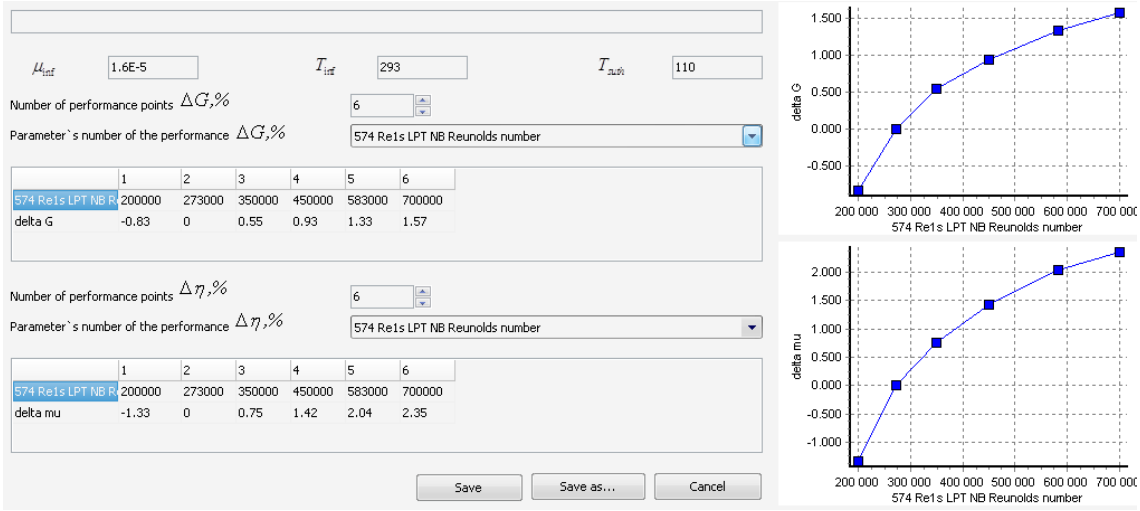


Figure 1

2. New version comprises an editor to input throat area of NVs of the first stage of LPT and chord of NVs of the first stage of LPT (path ***Initial data→Add files→Geometry. Geometrical parameters of an engine***) (fig. 2)

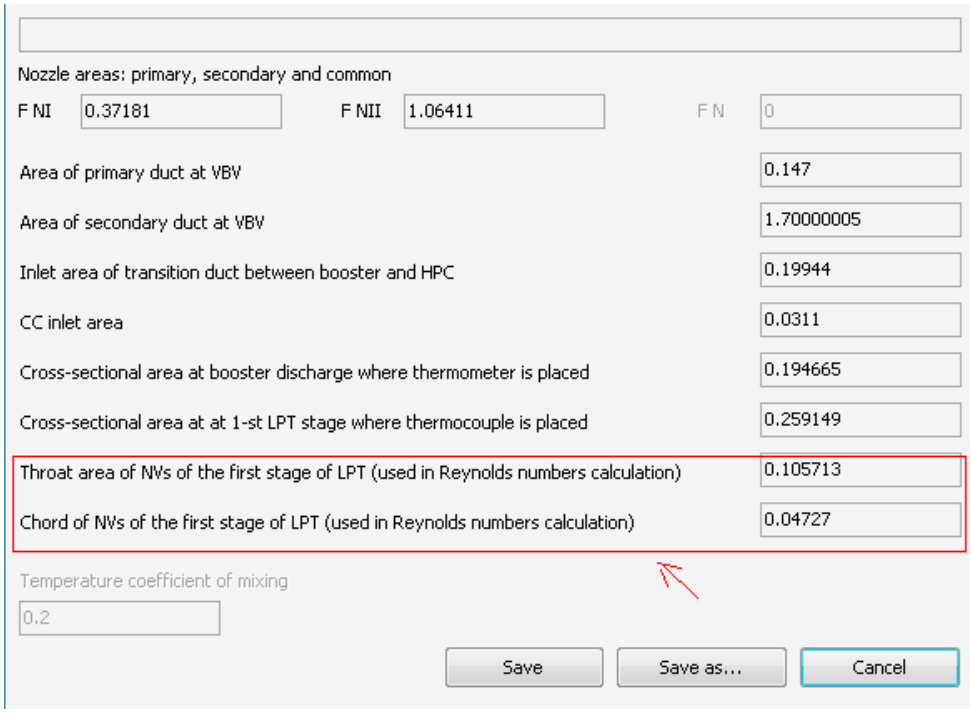


Figure 2

Equation Chapter 1 Section 1

What`s new in version 4.25.21.002

1. New version now has an editor to input information about air bleeding for spinner heating purposes (file ***Add files Dopotbor – Air bleeds and power take-offs for aircraft needs*** (fig. 3).

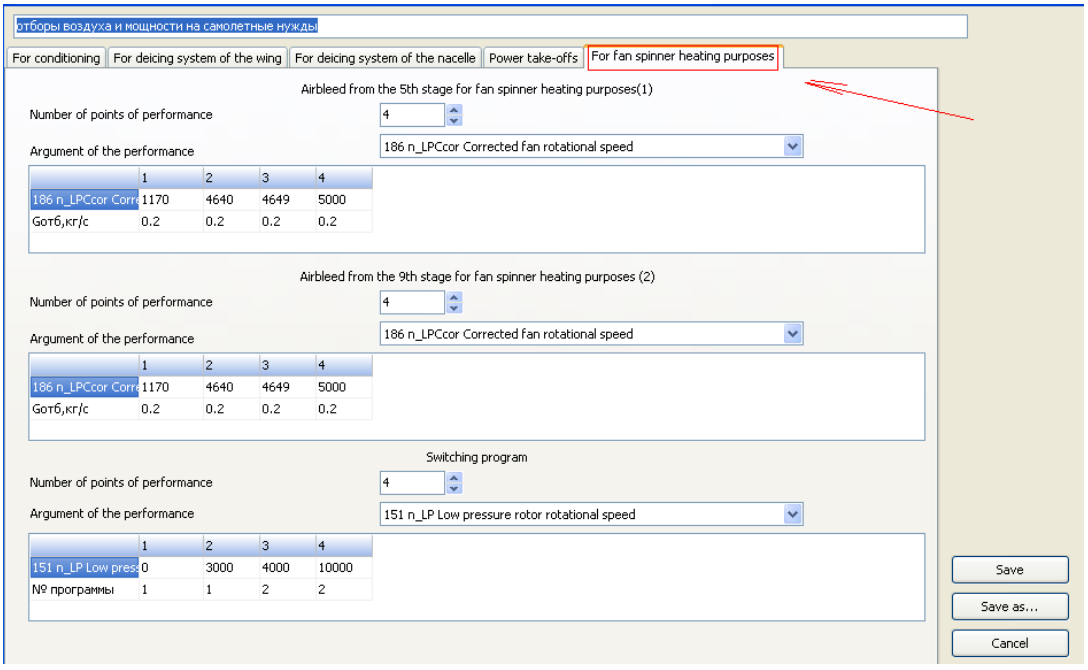


Figure 3

These bleeds can be set as a function of any parameter of gas path. Air bleeding from the fifth stage is calculated by the first program, so necessary data must be inputted in the field ***Bleeds from the fifth stage***, and from the ninth stage – by the second one, so necessary data must be inputted in the field ***Bleeds from the ninth stage***. Switching between bleeding programs happens according to settings set in corresponding field ***Switching program***. User also can set the argument of the switching program. To do this User must input 1 or 2 in line ***№ of program***.

To activate air bleeding it is necessary to set the flag that indicates switching it on and off. This must be done in menu ***Input parameters → Gas path → Input*** (fig. 4). To consider air bleeding for conditioning purposes User must add parameter № 591 I_hot and set value 1 (air is bleed) or 0 (air is not bled).

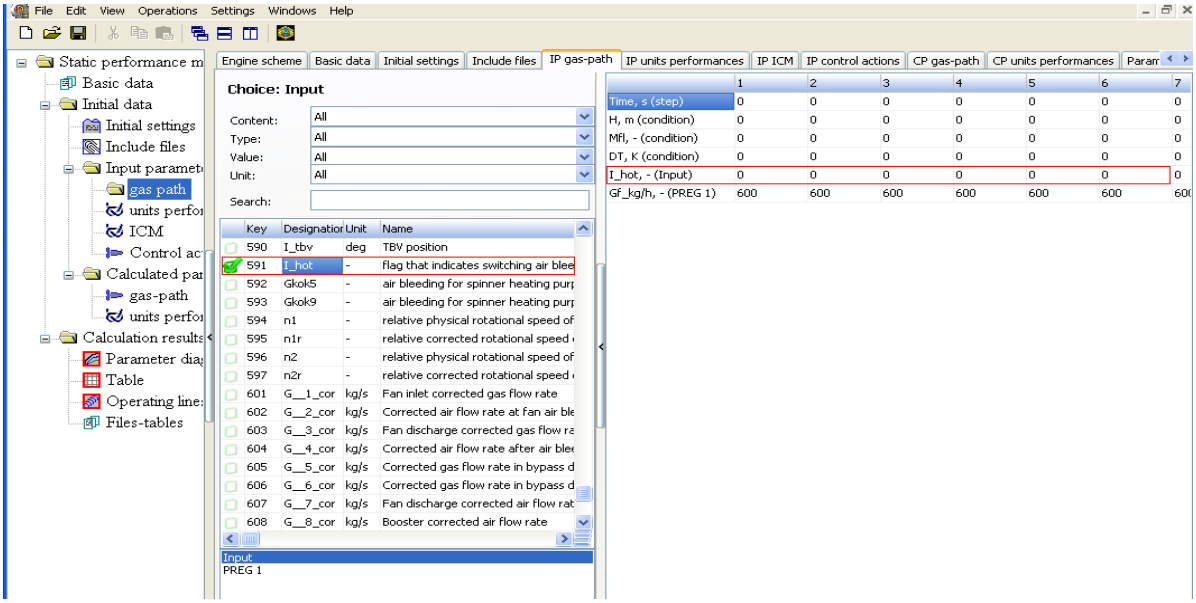


Figure 4

2. One independent variable T(2) (temperature at primary part of fan that considers air taps to heat the spinner) was added, because of new analysis. New analysis considers heating of fan spinner. Hence one line was added in files ***NEZP – Files of initial assumptions*** and ***Limit – Limits of independent variables*** (see fig. 5).

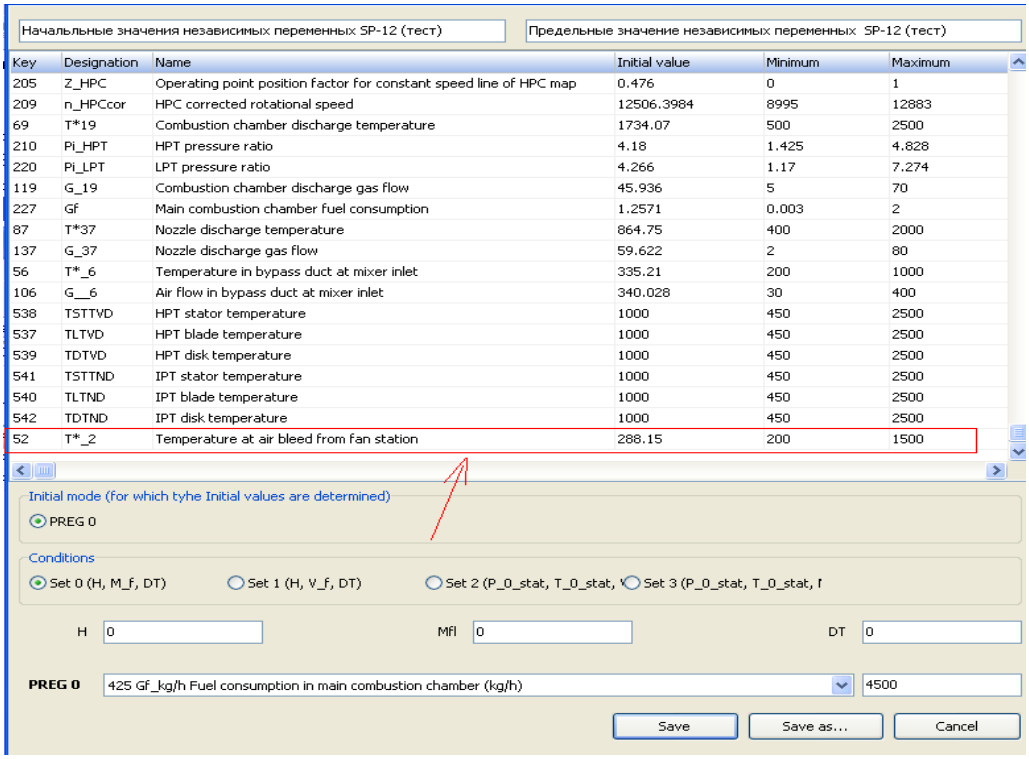


Figure 5

IMPORTANT: if you already have created files with initial assumptions and variation limits of independent variables you must update them by adding one line. Otherwise program will have input and processing errors.

3. Now User has the opportunity to input ambient conditions at which these assumptions were obtained (file ***NEZP – File with initial assumptions***) (see fig. 5).

4. To cope with the problem of the first point (when User specifies flight conditions different from standard) we added an option of forming file with initial assumptions automatically. To generate the file User must choose an item ***Initial data Initial settings Miscellaneous–Form file with initial assumptions*** (double click by left mouse button) (see fig. 6). Next, file ***Nezper-gen.dat*** will be automatically generated (to check or add this file use ***Initial data–Add files–NEZPFile with initial assumptions***). The program will output the initial assumptions of the latest successful analysis.

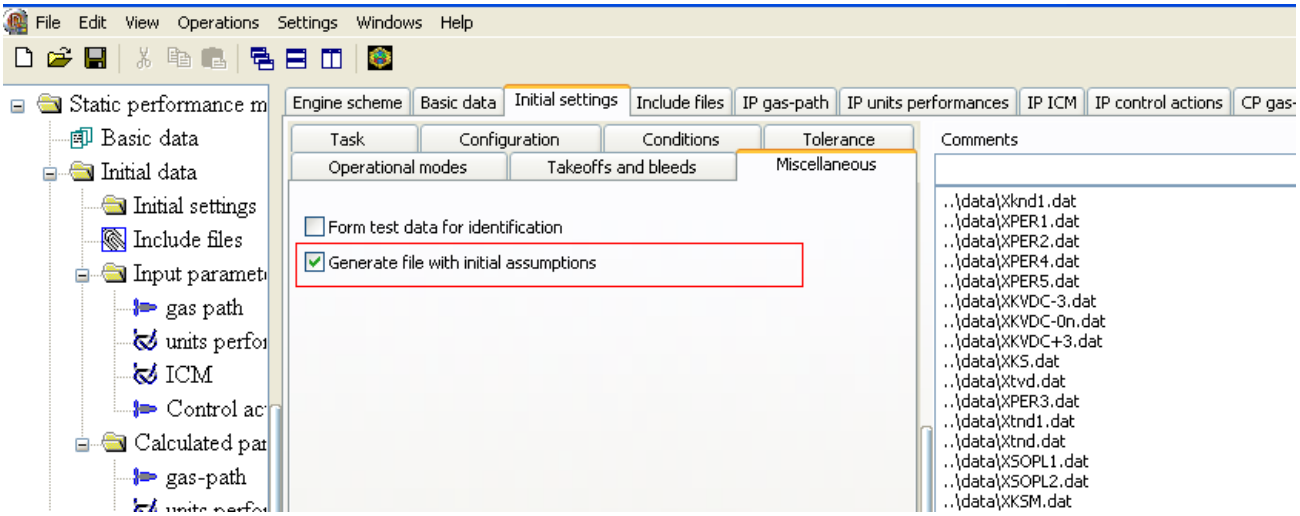


Figure 6

Example:

We need to simulate engine operation at the following mode: N1=3000 rpm, altitude 10000 m and flight speed 0,4 Mach. Let`s set these data in the program (***Input parameters–Gas path***) (fig. 7)

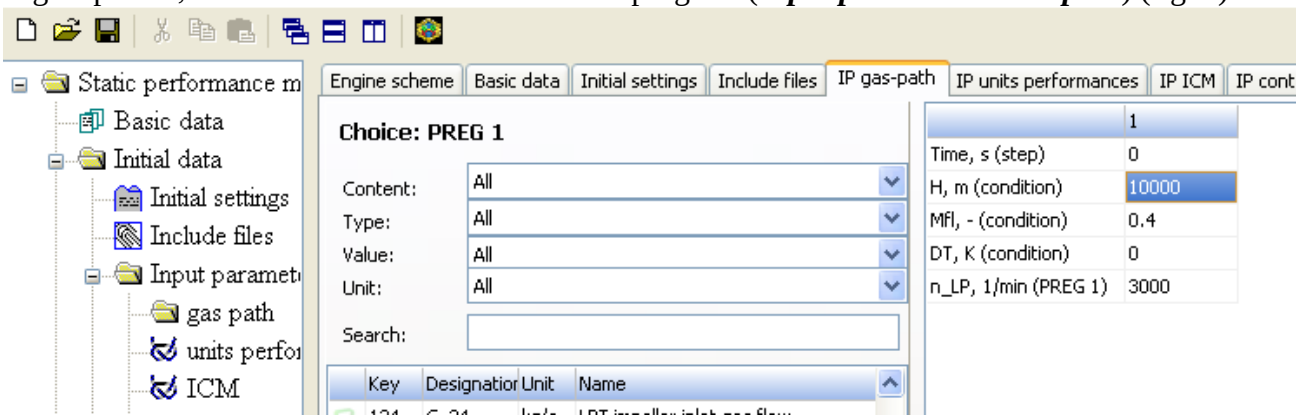


Figure 7

We choose file with initial assumptions nezper0.DAT. This file was formed for the mode H=0, M=0, Dt=0 и и Gt=4500 kg/h (editor is presented in fig. 8).

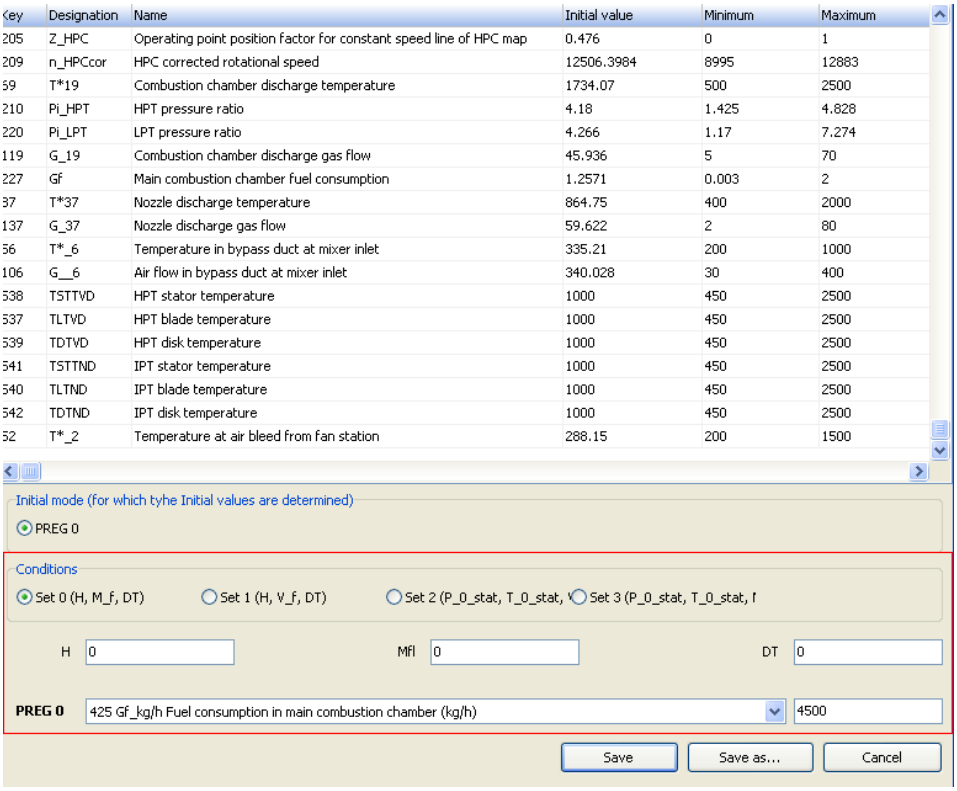


Figure 8

After we have started the simulation the message appears (fig. 9). That means that the analysis was not completed. But at the same time program is able to successfully complete this analysis but in standard ground conditions (N1=3000 rpm at altitude 0 m and flight speed 0 Machs).

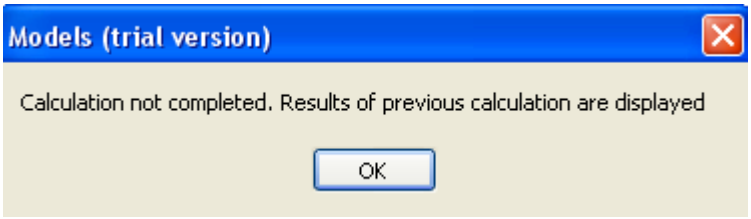


Figure 9

Now let`s gradually approach the needed mode (for some modes, starting from mode H=0, M=0, Dt=0, second mode H=5000, M=0,4, Dt=0, and next – H=10000, M=0,4, Dt=0, see fig. 10).

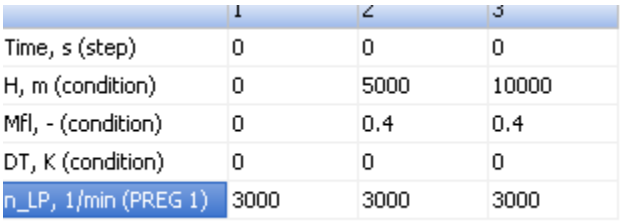


Рисунок 10

Next we choose ***Form file with initial assumptions*** as it is shown in fig. 6 and start an analysis. Program automatically generates file ***Nezper-gen.dat.*** We add it to the project (menu ***Initial data–Add files–NEZPFile with initial assumptions*** see fig. 11).

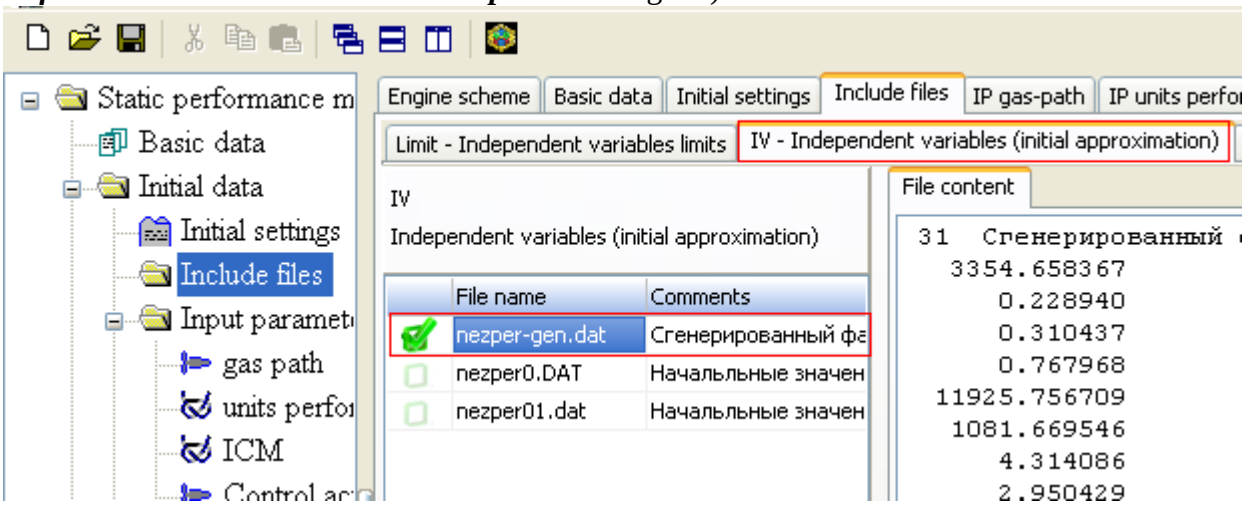


Figure 11

If to call an editor you will discover that initial assumptions were obtained for the mode N1=3000 rpm at an altitude 10000 m and flight speed 0,4 Machs (see fig. 12).

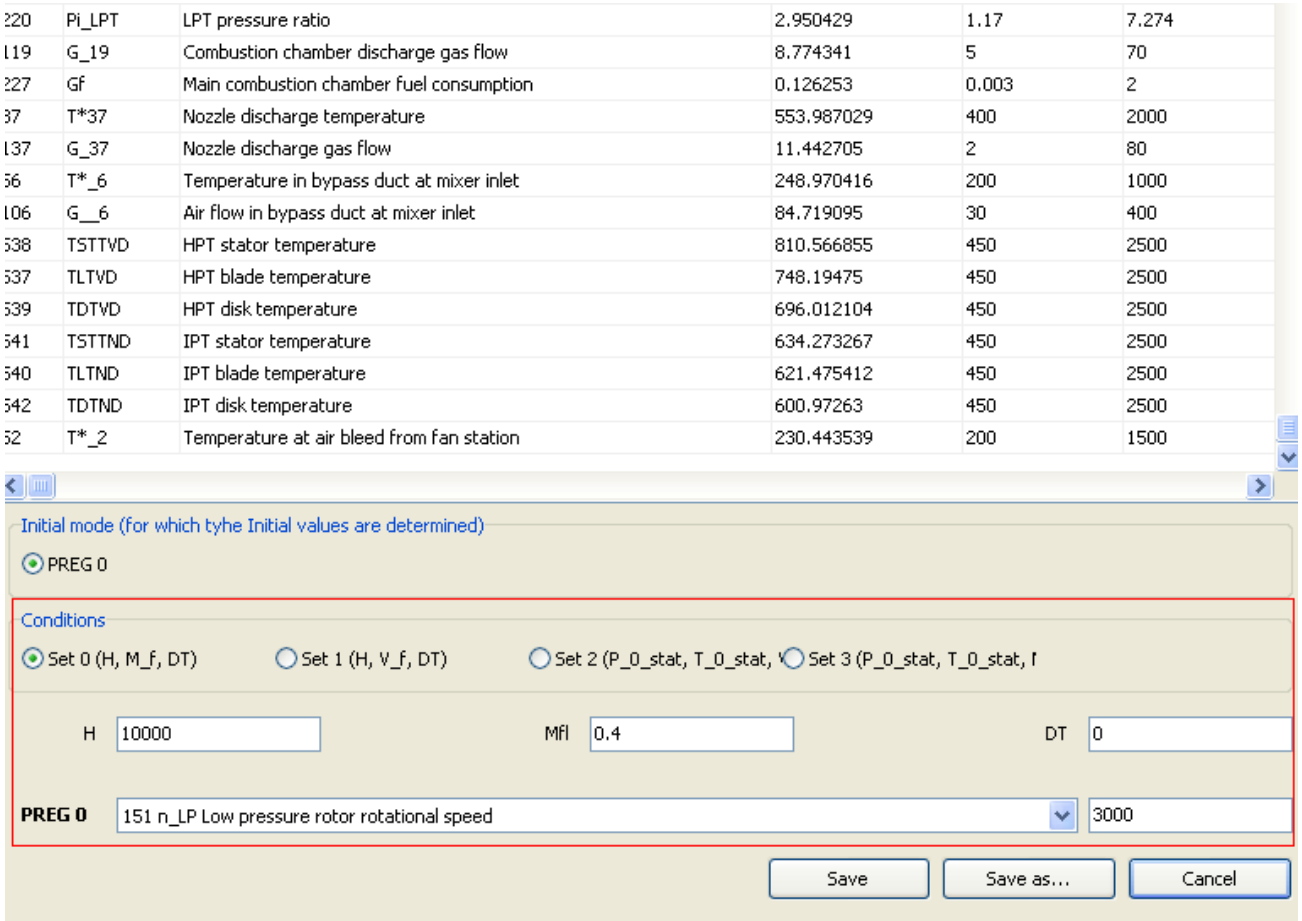


Figure 12

Now let`s again simulate engine operating mode N1=3000 rpm at altitude 10000 m and velocity 0,4 M (see. 7) (the mode which analysis failed previous attempt), but using initial assumptions from file Nezper-gen.dat. This time we have succeeded. Results of analysis are presented in fig. 13).

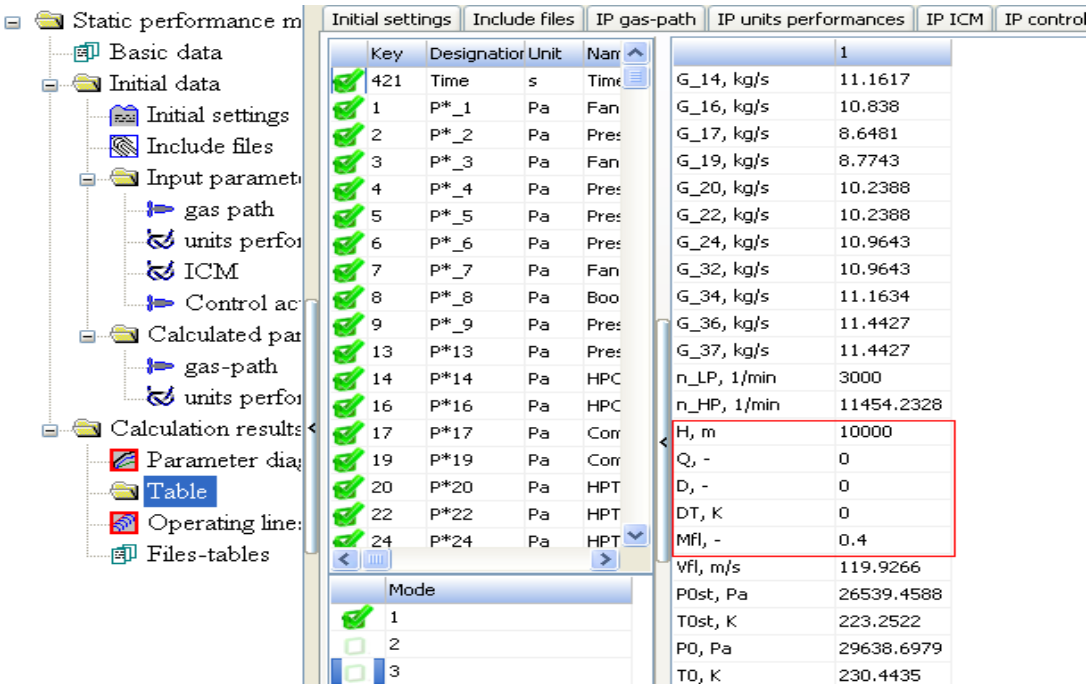


Рисунок 13

5. New version has an updated editor of mechanical properties (***Initial data–Add files–MEH–Mechanical properties and mechanization***), where values n1 100%, n1r 100%, n2 100% и n2r 100% are available for inputting.

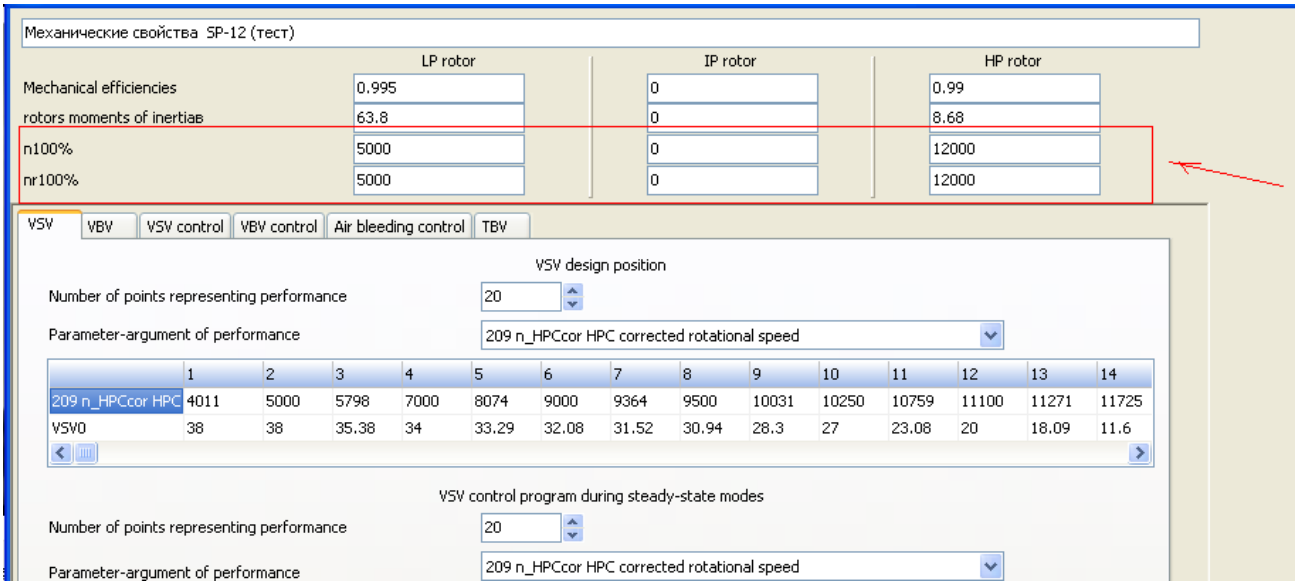


Figure 14

6. The following parameters were added

№ 591 I_hot – flag that indicates switching air bleeding for spinner heating on (1) / off (0);
№ 592 Gkok5 – air bleeding for spinner heating purposes from fifth stage;
№ 593 Gkok9 - air bleeding for spinner heating purposes from ninth stage;
№ 594 n1 – relative physical rotational speed of LP rotor;
№ 595 n1r - relative corrected rotational speed of LP rotor;
№ 596 n2 - relative physical rotational speed of HP rotor;
№ 597 n2r - relative corrected rotational speed of HP rotor;

What`s new in version 4.19.16.002

1. File with parameters of thermocouple was added to the program. To call the editor User must choose file ***TP – Thermocouple*** in menu ***Add files*** (Fig. **15**).

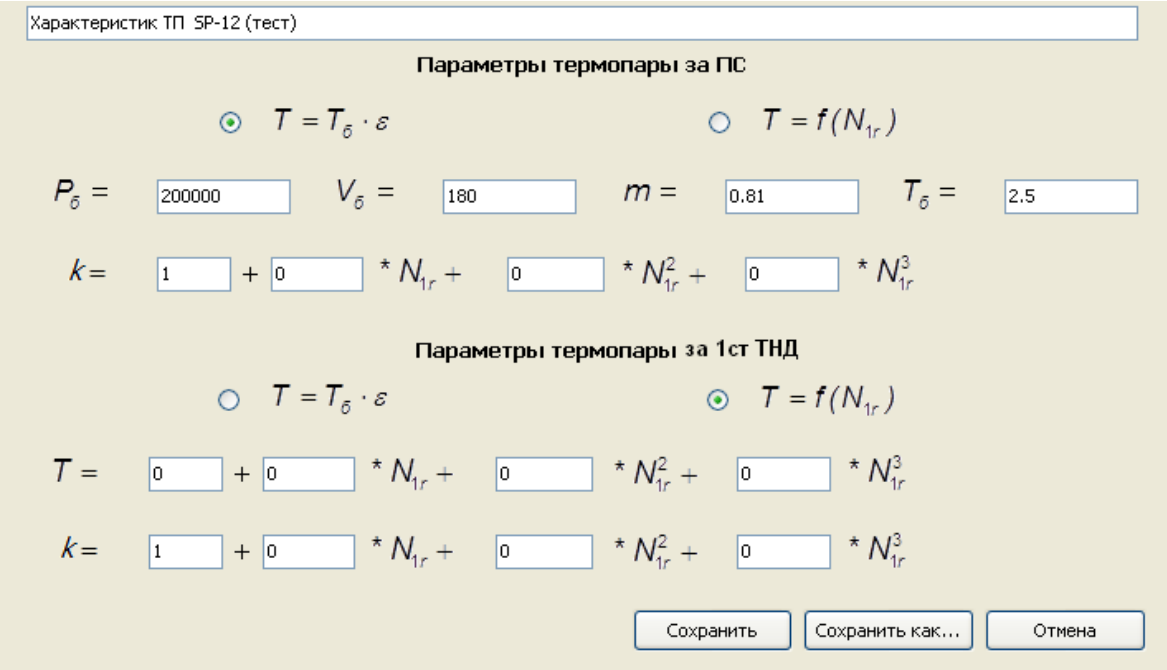


Figure 15

The editor has the fields for editing the parameters of thermocouples at boosters discharge and at first stage of LPT discharge. Time constant can be inputted in two ways:

1st way: $T = T_{base} \cdot \varepsilon$, where $\varepsilon = \left(\frac{p_{base} \cdot V_{base}}{p \cdot V} \right)^m$, here User must input the parameters p_{base}, V_{base}, m, T_{base} (e.g. parameters of thermocouple at boosters discharge are shown in figure 15);

2nd way: $T = f(N_{1r})$, where $f(N_{1r})$ is cubic polynomial that in its turn depends on N_{1r} $T = a_1 + a_2 \cdot N_{1r} + a_3 \cdot N_{1r}^2 + a_4 \cdot N_{1r}^3$. Here User must input coefficients a_1, a_2, a_3, a_4 (e.g. parameters of thermocouple at first stage of LPT discharge are shown in figure 15);

Gain coefficient is set in polynomial format, similar to time constant.

2. File to input the power take-offs and air bleeds for aircraft needs was added to the program. To call the editor User must choose file ***Dopotbor – power take-offs and air bleeds for aircraft needs***.

This editor consists of four tabs: ***For conditioning***, ***For deicing system of the wing***, ***For deicing system of the nacelle***, ***Power take-offs*** (Fig. **16**).

User is given an opportunity to set the absolute air bleeds from fifth and ninth stage separately. Bleeds and take-offs can be set to be dependent from any parameter along the gas path. Bleeds and take-offs must be specified in field ***Bleed from the fifth stage*** (program №1). Bleeds and take-offs must be specified in field ***Bleed from the ninth stage*** (program №2). Switching between programs can be set in corresponding field ***Switching program***. User also can choose parameter that will be an argument of switching program. To choose active bleed program User must input value 1 or 2 in the field № of program.

To take these bleeds and take-offs into account User must activate the switching flag (switched on or off) in menu item ***Input parameters → Gas path → Input*** (fig. **16**).

To activate air bleeding for conditioning purposes User must add parameter ***I_Gcond*** (№ 584) and specify its value. In case "1" is specified then air is bleed, otherwise ("0") – air is not bleed.

To activate air bleeding for deicing system of the nacelle User must add parameter ***I_GAIM*** (№ 586) and specify its value. In case "1" is specified then air is bleed, otherwise ("0") – air is not bleed.

To activate power take-offs User must add parameter ***I_Na*** (№ 587) and specify its value. In case "1" is specified then power is taken off, otherwise ("0") – power take-offs are absent.

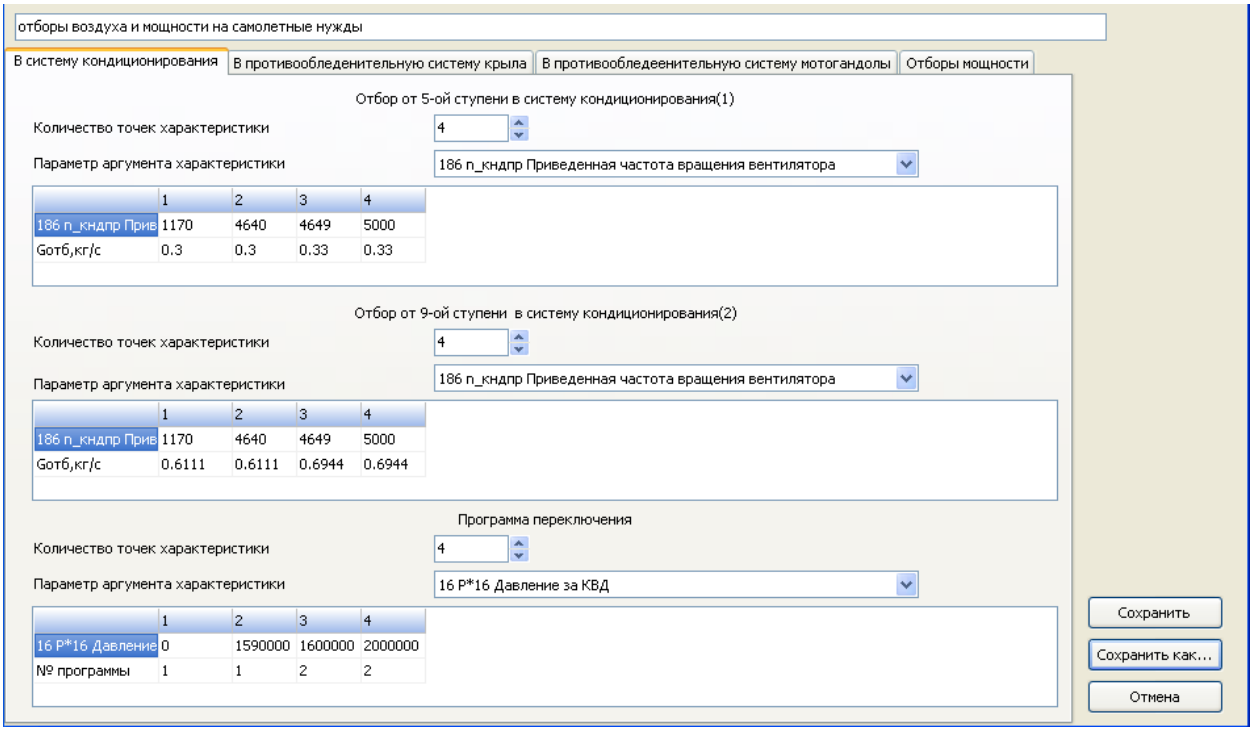


Figure 16

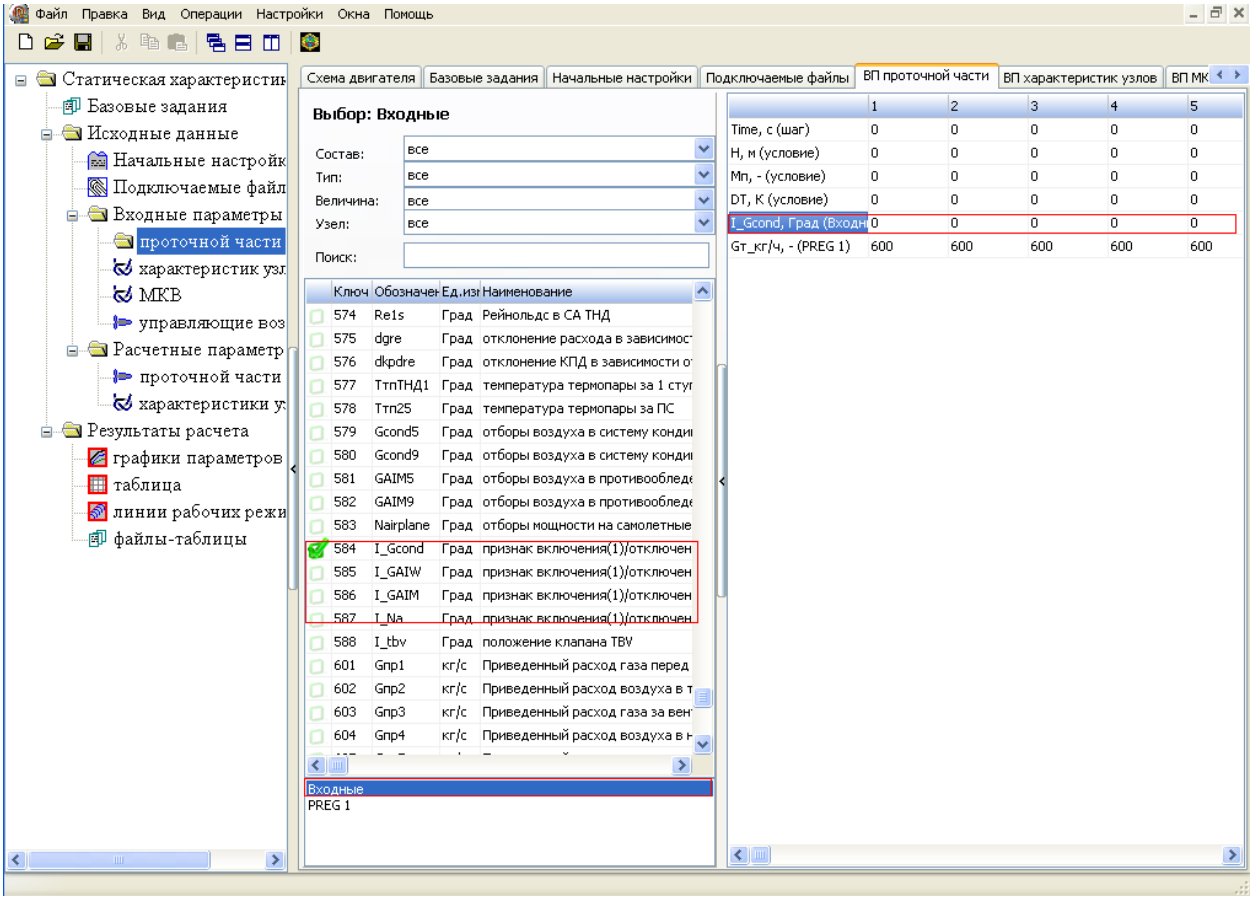


Figure 17

3. I have deleted the opportunity to input the initial value and variation limits of pressure ratio of the LPT first stage in the editors NEZP and Limit.

4. To input the parameters of the static governor that controls given value of corrected rotational speed User must choose ***Relation type*** => ***Analytical relation*** (see fig. 18). Value of corrected rotational speed is determined according to the following equation:

$$N_{1ry\ cor} = N_{1ry\ stand} + \Delta N_{1r} \qquad (1.1)$$

where $N_{1ry\ stand}$ is determined from the table 1, ΔN_{1r} is determined from the table 2.

Table 1

PLA(°)	130	125	120	115	110	≤55
$N_{1ry\ stand}$ (rpm)	4749	4749	4649	4533	4358	1170

Table 2

P_0(kPa)	108.7	101.325	89.87	38.26	30.75	25.4	19.3	19.3	0
ΔN_{1r} (rpm)	0	0	118	611	717	780	780	780	780

User must specify the relation format "***Tabular format***" in the field ***Parameter Y1*** and input values from Table 1. User must do the same with the field ***Parameter Y2*** but input values from Table 2. The argument of the relations is an angle of thrust lever (parameter №422). Field ***Parameter Y3*** can be filled with arbitrary values because these values are not considered during the simulation.

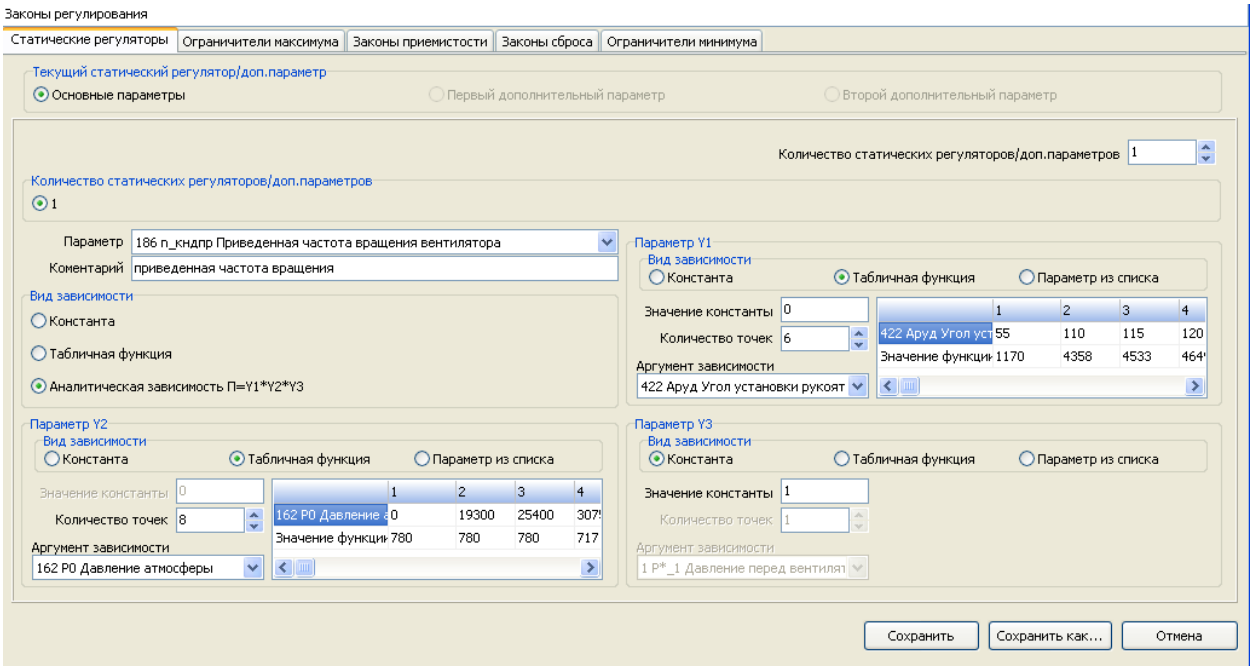


Figure 18

IMPORTANT. These format of the governor ($N_{1ry\ cor} = N_{1ry\ stand} + \Delta N_{1r}$) is applicable ONLY for the first static governor. If there are some of them then parameters of the rest governors are calculated by equation $\Pi = Y1 \cdot Y2 \cdot Y3$.

5. To input the limits of corrected rotational speeds of low pressure rotor ($N_{1ry\ cor}$) User must choose ***Relation format*** => ***Analytical relation*** for the first parameter in the editor ***Maximum limiter*** (see fig. 19).

Table 3

For PLAs bigger than 125 degrees	T_2 (K)	373	323	303	279.8	271.1	250
	Limiting value of n_{1R}, (rpm)	3431	4365	4749	5142	5300	5300
For PLA = 120 degrees	T_2 (K)	373	323	298.15	262.96	258.3	250
	Limiting value of n_{1R} (rpm)	3300	4156	4649	5234	5300	5300
For PLAs less	T_2 (K)	353	293.2	254.6			

than 115 degrees	Limiting value of n_{1R} (rpm)	3388	4530	5300			

User must specify the relation format "***Tabular format***" in the field ***Parameter Y1*** and input values from Table 3 (values that correspond to PLAs less than 115 degrees). User must do the same with the field ***Parameter Y2*** but input other values from Table 3 (values that correspond to PLA = 120 degrees). The argument of both relations is temperature at engine inlet (parameter №163). User must also specify the relation format (***Relation format*** => ***Tabular relation***) and input values from the Table 3 (values that correspond to PLAs bigger than 125 degrees).

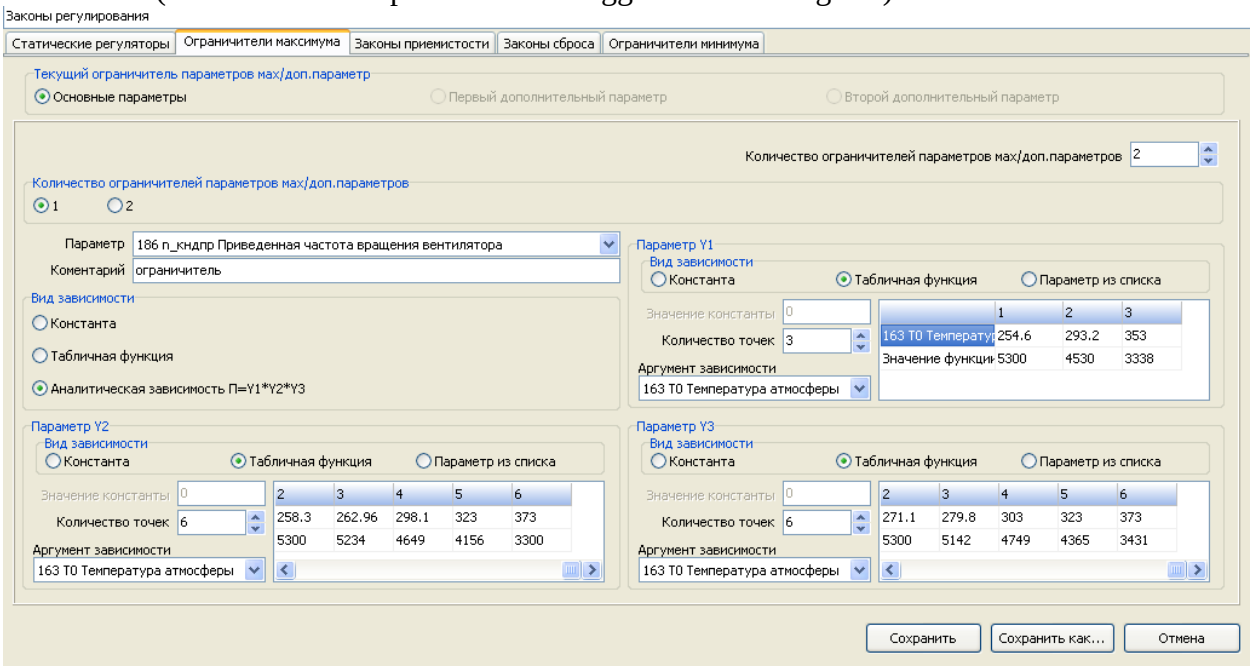


Figure 19

IMPORTANT. These format to input parameters of the governor is applicable ONLY for the first maximum value limiter. If there are some of them then parameters of the rest governors are calculated according to standard relations.

6. Idle mode is determined by the inlet corrected rotational speed of HPC (N_{2r}**).** Static governor is based on N_{1r}. Hence to limit the parameters at idle mode User must input the limiting value of N_{2r} in the editor ***Minimum limiters*** (fig. 20).

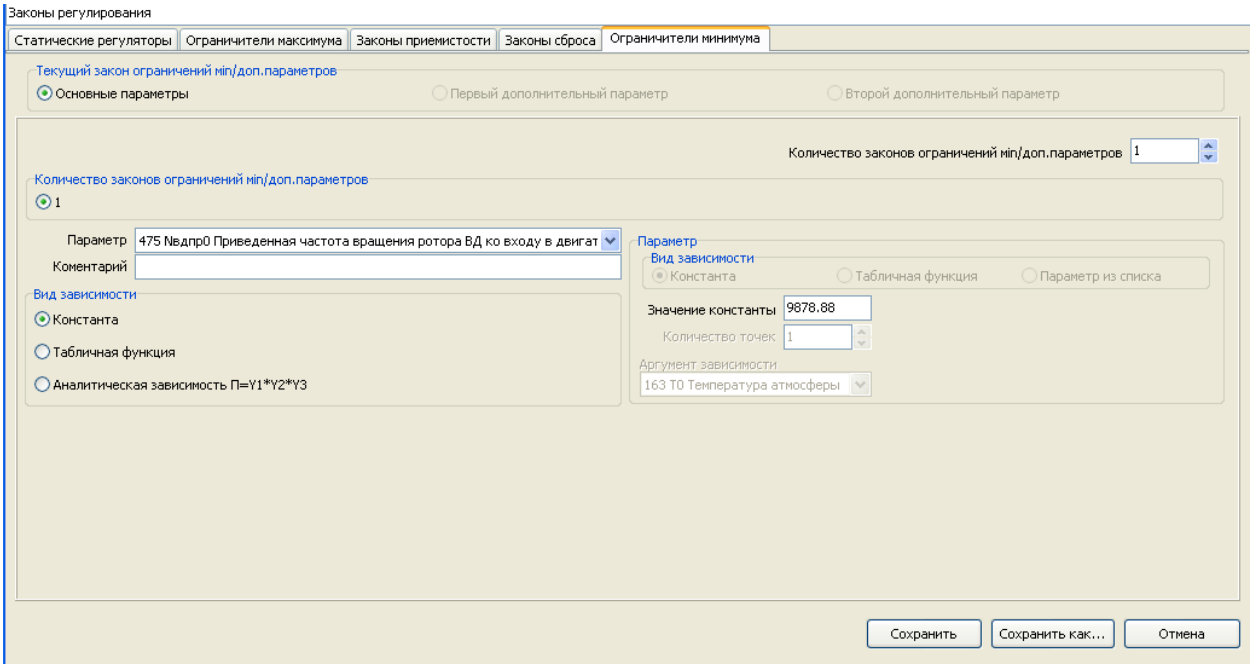


Figure 20

7. The following parameters are now available as input actions:
VBV position (№565 in general list of parameters)
VSV position (№569 in general list of parameters)
TBV position (№588 in general list of parameters)
bleeds and take-offs control program (№549 in general list of parameters)
Hence, LDM with this parameters is as follows:

$$\dot{n}_i = a_{i1}\Delta n_1 + a_{i2}\Delta n_2 + b_{i1}\Delta G_f + b_{i2}\Delta VBV + b_{i3}\Delta VSV + b_{i4}i_{TBV} + b_{i5}i_{cor\ bleed} \quad (1.2)$$

$$\Delta Y_j = c_{j1}\Delta n_1 + c_{j2}\Delta n_2 + d_{j1}\Delta G_f + d_{j2}\Delta VBV + d_{j3}\Delta VSV + d_{j4}i_{TBV} + d_{j5}i_{cor\ bleed} \quad (1.3)$$

where ΔVBV and ΔVSV are angular deviations of VBV and VSV from set in the control program.

So, to get the corresponding coefficients of matrices B and D User must specify the absolute deviations from set in the control program using menu item ***Input parameters =>Control actions***.

Calculation of influence coefficients of matrices B and D for parameters i_{TBV} - position of TBV valve and $i_{cor\ bleed}$ - program to control air bleeds and taps. These parameters are discrete, so there is no need to specify their deviations in menu item ***Input parameters =>Control actions***. If i_{TBV} was chosen as control action then subroutine forms two coefficients *b* for each rotor and two coefficients в for each supervision (observation) parameter (for open and closed position).

If there is necessary to simulate opening or closing the valve at any engine operating mode, User must manually input coefficients from the column I_tbv_CLOSE (valve is closing) or from the column I_tbv_OPEN (valve is opening). E.g. if there are zeroes in the column I_tbv_CLOSE then the valve is modeled to be closed. This indicator makes no influence on parameters.

All the considerations are valid for the parameter $i_{cor\ bleed}$. The only difference is that there are three discrete positions: bleeding from the 5th stage (protb_5 – name of the column in matrices *b* and *d*), bleeding from the 9th stage (protb_9 – name of the column in matrices *b* and *d*) and bleeding from both 5th and 9th stages (protb_5+9 – name of the column in matrices *b* and *d*). Consequently,

three columns appear for this parameter. Parameters $i_{cor\ bleed}$ must be modeled analogically to parameter i_{TBV}.

8. The editor to set the impact of clearance on the efficiency of HPT was added. To call the editor User must choose "DKPDTVD – The performance of efficiency of deviation of HPT efficiency due to clearance" in the add files. Pressing the Editing button calls the standard editor.

What`s new in version 4.16.12.001

1. The error that appeared when saving the performances of compressors and turbines using the graphics editor is now fixed.
2. Search option (***Search*** is available for designations of parameters or their names) was added for the menu ***Input parameters – gas-path***. The search is case-insensitive. The request must be identical to designation or name of parameter.

What`s new in version 4.15.12.000

1. The format of the file with compressor and turbine performances was changed. The changes regard to the line with corrected rotational speed and number of points of constant speed line. **IMPORTANT: The performances with more than 10 points of one constant speed line, which were created in versions 2.x.x – 3.x.x will not work properly in the new version. To use them in program versions higher than 4.0.0 User must edit them in any text editor (add space between corrected rotational speed and number of points of constant speed line).**
   Example: "10794.6 rpm" constant speed line of compressor performance is formed by 10 points. Consequently, the line in lower version is "10794.610". To adopt this performance to the new one version User must add at least one space between value of corrected rotational speed and number of points of constant speed line. The new line will have the following format – "10794.6 10".
2. The editor of geometrical performances of the engine was added.
3. Two more editors were added for inputting different schemes of air bleeds and taps and power take-offs and supplies.
4. The editor for inputting control laws was added.
5. The editor for inputting information needed for calculating the radial displacements of HPT casing was changed.
6. The editor for inputting information needed for calculating the radial displacements of HPT rotor was changed.
7. The editor for inputting information needed for calculating the radial displacements of HPT blade was changed.
8. The algorithm that calculates radial displacements of HPT rotor was changed.

9. The algorithm that calculates radial displacements of HPT casing was changed.
10. The algorithm that calculates radial displacements of HPT blade was changed.
11. The editor for inputting the control settings of VBV, VSV, TBV, air bleeds and also for inputting their performances.
12. Now program can calculate the temperature after boosters and at the discharge of first stage of LPT. Temperatures are measured by thermocouple.
13. Now program can save the performances of components taking into account the state parameters.
14. Additional fields were added in the editors of performances to input the information used for identification. The identification is run by introducing coefficients that express the turning of the performance around the given point.

What`s new in version 3.0.4.000

1. The error that appeared when using DGk(8) was fixed for all engine types.
2. Coefficients k_π now are calculated in the similar way for performances of compressors and turbines $k_\pi = \frac{\pi_{T\ cor} - 1}{\pi_{T\ perf} - 1}$
3. The error of visualizing operating line at performances maps was fixed.
4. The formatted input of correction factors for performances of turbines and compressors was canceled.

What`s new in version 3.0.0.000

In program modules

1. Size of "bleeds-taps" matrix was increased up to 10 bleed stations (3 stations in LPC, 1 station at boosters discharge, 3 stations in IPC, 3 stations in HPC) and 11 tap stations (stations available for air bleeding are given in subsection 6.1.2.8 of the Manual).
2. New version has sign, which identifies the successful completion of calculation (File rez.dat is formed).
3. This version has improved algorithm that calculates temperature of air at bleed stations (subsection 1.1.2.14).
4. In this version gas-path straight-though calculation does not take into account heat exchange between gas (air) and HPT structural elements.
5. Now User can input influence coefficient and blade length in files TRANSBLV, TRANSBLS and TRANSBLN (subsection 6.1.2.12).
6. New version has control coefficients that allow choosing additional dynamic factors for dynamics calculation.

7. While customizing performance of combustion chamber, User can specify fuel enthalpy at combustion chamber inlet i_0, lower caloric value of used fuel H_u and stoichiometric ratio of used fuel *L*(see subsection 6.1.2.4 of the Manual).
8. This version has control coefficient that identifies the form of HPC, IPC, LPC and boosters performances (the sense of the coefficient is in subsection 6.1.2.2 of the Manual). An algorithm of calculating design point at the performance map was correspondingly changed.
9. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP).
10. This version has two new studies – Identification and Calculation of QLDM coefficients.
11. Variables in file Namepar.txt have a new order (starting from 291th parameter), which is presented in Appendix E of the Manual.
12. Bugs in calculating temperature of HPT blade (TLTVD), temperature of HPT disk (TDTVD), temperature of IPT blade (TLTSD), temperature of IPT disk (TDTSD), temperature of LPT blade (TLTND), temperature of LPT disk (TDTND) were fixed.
13. Calculating of QLDM coefficients was improved by taking into account additional dynamic factors.
14. When outputting the matrix of QLDM coefficients, program outputs name of mode parameter instead of its number.

GTE-1

1. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of nozzle, heat exchange between gas and structural elements of HPT and FT, changing of radial clearances in HPT and FT. Because of this reason the number of independent variables was increased from 12 up to 18 (the same was done for output windows of independent variables and residuals).
2. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.15 of the Manual).
3. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

4. Number of iterations in identification was increased from 5 to 10.

GTE-2

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.
2. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of nozzle, heat exchange between gas and structural elements of HPT, IPT and FT, changing of radial clearances in HPT, IPT and FT. Because of this reason the number of independent variables was increased from 17 up to 26 (the same was done for output windows of independent variables and residuals).
3. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.16 of the Manual).
4. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTS – transient performance of IPT stator, TRANSBLS – transient performance of IPT blade, TRANSDS – transient performance of IPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TJE-1

1. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.9 of the Manual).
2. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TJE-2

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.
2. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of nozzle, heat exchange between gas and structural elements of HPT and IPT, changing of radial clearances in HPT and IPT. Because of this reason the number of independent

variables was increased from 17 up to 21 (the same was done for output windows of independent variables and residuals).
3. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.10 of the Manual).
4. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTS – transient performance of IPT stator, TRANSBLS – transient performance of IPT blade, TRANSDS – transient performance of IPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TFE-2

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.
2. In this version there is a new option to model engines with boosters, and consequently, there is an opportunity to set the performance of boosters in including files.
3. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of secondary nozzle, volume of primary nozzle, heat exchange between gas and structural elements of HPT and LPT, changing of radial clearances in HPT and LPT. Because of this reason the number of independent variables was increased from 18 up to 25 (the same was done for output windows of independent variables and residuals).
4. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.11 of the Manual).
5. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TFE-2mix

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.
2. In this version there is a new option to model engines with boosters, and consequently, there is an opportunity to set the performance of boosters in including files.

3. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of secondary nozzle, volume of primary nozzle, heat exchange between gas and structural elements of HPT and LPT, changing of radial clearances in HPT and LPT. Because of this reason the number of independent variables was increased from 13 up to 28 (the same was done for output windows of independent variables and residuals).
4. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.13 of the Manual).
5. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TFE-3

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.
2. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of secondary nozzle, volume of primary nozzle, heat exchange between gas and structural elements of HPT, IPT and LPT, changing of radial clearances in HPT, IPT and LPT. Because of this reason the number of independent variables was increased from 17 up to 32 (the same was done for output windows of independent variables and residuals).
3. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.12 of the Manual).
4. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTS – transient performance of IPT stator, TRANSBLS – transient performance of IPT blade, TRANSDS – transient performance of IPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

TFE-3mix

1. Parameters in file Namepar1.txt, used for calculating additional dynamic factors, are now available for printing.

2. This engine type now can take into account the following dynamic factors: compressibility of working substance in combustion chamber, volume of secondary nozzle, volume of common nozzle, heat exchange between gas and structural elements of HPT, IPT and LPT, changing of radial clearances in HPT, IPT and LPT. Because of this reason the number of independent variables was increased from 18 up to 33 (the same was done for output windows of independent variables and residuals).
3. Order of parameters in the file with variation limits of independent variables (Limit) is the same as in the file with initial approximations of independent variables (NEZP) (see subsection 6.1.2.10 and Table 6.14of the Manual).
4. There are new include files for this engine type: nezp – initial approximations of independent variables, tepl – heat exchange parameters (subsection 6.1.2.11 of the Manual), TRANSSTV – transient performance of HPT stator, TRANSBLV – transient performance of HPT blade, TRANSDV – transient performance of HPT disk, TRANSSTS – transient performance of IPT stator, TRANSBLS – transient performance of IPT blade, TRANSDS – transient performance of IPT disk, TRANSSTN – transient performance of LPT stator, TRANSBLN – transient performance of LPT blade, TRANSDN – transient performance of LPT disk, volume – characteristic volumes (subsection 6.1.2.14 of the Manual).

In interface

1. It is forbidden to edit performances in the right text box.
2. Program is complemented with the editor of intake (subsection 6.1.2.1).
3. Program is complemented with the editor of compressors (subsection 6.1.2.2). Now it is possible to add corrective and control coefficients.
4. Program is complemented with the editor of transition ducts (subsection 6.1.2.3).
5. Program is complemented with the editor of combustion chamber (subsection 6.1.2.4).
6. Program is complemented with the editor of turbines (subsection 6.1.2.5).
7. Program is complemented with the editor of nozzles (subsection 6.1.2.6).
8. Program is complemented with the editor of geometrical parameters of the engine (subsection 6.1.2.7).
9. Program is complemented with the editor air bleeds and taps, power take-offs and supplies (subsection 6.1.2.8).
10. Program is complemented with the editor of mechanical properties and compressor mechanization (subsection 6.1.2.9).
11. Program is complemented with the editor of initial approximations and variation limits of independent variables (subsection 6.1.2.10). Now program checks if the initial approximation lies within the limits, and in case it is not true - forms the corresponding message.
12. Program is complemented with the editor of heat exchange parameters (subsection 6.1.2.11).

13. Program is complemented with the editor of transient performances of structural elements (subsection 6.1.2.12).
14. Program is complemented with the editor of flow splitter (subsection 6.1.2.13).
15. Program is complemented with the editor of parameters that describe characteristic volumes (subsection 6.1.2.14).
16. Editing can be performed by pasting one value or by some values from tables, which were copied from EXCEL.
17. Performances, set in tables of the editor, are displayed in graphical form.
18. Dynamics and QLDM coefficients calculations of TJE-2, TFE-2, TFE-2mix, TFE-3, TFE-3mix, GTE-1, GTE-2 now can take into account additional dynamic factors.
19. Now program can calculate corrected mass flow at each station for all engine types.
20. There is a self-testing in the program to check presence of all necessary files. If not all files are included, then program forms the corresponding message.
21. The list of parameters, available for choosing them as control actions, is limited in the study “QLDM coefficients calculation”.

In the Manual

1. Manual is complemented with the description of calculating parameters at bleed stations (subroutine TAU), see subsection 1.1.2.14.
2. Order of description was changed, now static performance goes first, and dynamics – next.
3. Manual is complemented with the calling scheme of subroutines used to calculate static performances of three-spool TFE with flows mixing (fig. 1.3).
4. Manual is complemented with the equation to calculate time constant of volume (subsection 1.3.2).
5. Manual is complemented with the equation to calculated time constant of structural element (subsection 1.3.3).
6. Manual is complemented the calling scheme of subroutines used to calculate dynamics of three-spool TFE with flows mixing (fig. 1.7).
7. Manual is complemented with the description of QLDM structure (see subsection 1.4.1).
8. Manual is complemented with the description of how to form static model (see subsection 1.4.2).
9. Manual is complemented with different variants to form QLDM (figs. 1.9-1.11).
10. Manual is complemented with the description of method to calculate QLDM coefficients (subsection 1.4.3).
11. Manual is complemented the calling scheme of subroutines used to calculate QLDM coefficients of three-spool TFE with flows mixing (fig. 1.12).
12. Manual is complemented the description of method to synthesize multimode QLDM (subsection 1.4.4).

13. Manual is complemented with the description of the list of included files (fig 6.11).
14. Manual is complemented with the list of performances, which are used in models (table 6.1).
15. Manual is complemented with the description of intake editor (subsection 6.1.2.1).
16. Manual is complemented with the description of compressor editor (subsection 6.1.2.2).
17. Manual is complemented with the description of editor of transition duct (subsection 6.1.2.3).
18. Manual is complemented with the description of editor of combustion chamber (subsection 6.1.2.4).
19. Manual is complemented with the description of turbine editor (subsection 6.1.2.5).
20. Manual is complemented with the description of nozzles editor (subsection 6.1.2.6).
21. Manual is complemented with the description of the editor to customize geometrical parameters of an engine (subsection 6.1.2.7).
22. Manual is complemented with the description of parameters, available to customize geometrical parameters of all engine types (table 6.2).
23. Manual is complemented with the description of the editor to customize air bleeds and taps, power take-offs and supplies (subsection 6.1.2.8).
24. Manual is complemented with the description of stations, which are available for air bleeds and taps, power take-offs and supplies, for all engine types (tables 6.3-6.8).
25. Manual is complemented with the description of the editor to customize mechanical properties and compressors mechanization (subsection 6.1.2.9).
26. Manual is complemented with the description of the editor to set initial approximations of independent variables (subsection 6.1.2.10).
27. Manual is complemented with the description of the list of independent variables for all engine types (tables 6.9 – 6.16).
28. Manual is complemented with recommendations about setting variation limits of independent variables (subsection 6.1.2.10).
29. Manual is complemented with the description of the editor to customize parameters of heat exchange (subsection 6.1.2.11).
30. Manual is complemented with the description of the editor to customize transient performances (subsection 6.1.2.12).
31. Manual is complemented with the description of the editor to customize parameters of flow splitter (subsection 6.1.2.13).
32. Manual is complemented with the description of the editor to customize parameters of characteristic volumes (subsection 6.1.2.14).
33. Manual is complemented with the description of how to configure the studies (straight-through calculation – subsection 6.3.1, static performance – subsection 6.3.2,

identification – subsection 6.3.3, dynamics – subsection 6.3.4, QLDM coefficients calculation – subsection 6.3.5).
34. Manual is complemented with references.
35. Manual is complemented with the description of file content (file with transient performances of stator and disk, see Appendix A.15).
36. Manual is complemented with the description of file content (file with transient performances of blade, see Appendix A.14).
37. Manual is complemented with the description of file content (file with parameters of characteristic volumes, see Appendix A.16).
38. Manual is complemented with calling schemes of subroutines for all engine types (see Appendix B).
39. Manual is complemented with error messages, which may appear during calculation (see Appendix D).
40. Manual is complemented with the list of additional parameters, used for each engine type (Appendix E).
41. Manual is complemented with the description of main variables, used in all program modules (Appendix F).